**COURSE TITLE : BIOMEDICAL INSTRUMENTS**
**COURSE CODE : 6213**
**COURSE CATEGORY : A**
**PERIODS/WEEK : 4**
**PERIODS/SEMESTER: 60**
**CREDITS : 4**

**<u>TIME SCHEDULE</u>**

| Module | Topics | Periods |
|---|---|---|
| 1 | Physiological System of Human Body and Blood Pressure Measurement | 15 |
| 2 | ECG, EMG and EEG Measurement | 15 |
| 3 | Therapeutic and Analytic Instruments | 15 |
| 4 | Imaging Systems, Bio telemetry & Safety Precautions | 15 |
| **Total** | | **60** |

**<u>COURSE OUTCOME</u>**

| Module | G.O | On completion of the study of this Module the student will be able |
|---|---|---|
| 1 | 1 | To understand the physiological systems of human body |
| | 2 | To Comprehend the working of different physiological transducers |
| | 3 | To Understand Blood Pressure measurement |
| | 4 | To know Blood flow meters |
| 2 | 1 | To know ECG measurement |
| | 2 | To comprehend EMG measurement |
| | 3 | To Understand EEG measurement |
| 3 | 1 | To Comprehend different types of therapeutic instruments |
| | 2 | To Understand blood cell counting technique. |
| 4 | 1 | To Understand different imaging techniques in medical field |
| | 2 | To know biotelemetry system |
| | 3 | To Comprehend safety precautions in biomedical field |

On completion of the study the student will be able

## <u>MODULE I</u> PHYSIOLOGICAL SYSTEM OF HUMAN BODY AND BLOOD PRESSURE MEASUREMENT

**1.1.0 To understand the physiological systems of human body**
1.1.1 To define bio-electricity.
1.1.2 To explain resting and action potentials of a cell.
1.1.3 To describe Sodium pump and transmission of impulses.

**1.2.0 To understand the working of physiological transducers.**
1.2.1 To explain the criteria for selecting biomedical transducers.
1.2.2 To explain the working principle of optical fiber temperature sensors.
1.2.3 To describe the working principle of photo electric pulse transducers.
1.2.4 To describe the principle of piezo-electric arterial pulse receptor.
1.2.5 To explain the working principle of strain gauge type respiration sensor.

**1.3.0 To Understand Blood Pressure measurement.**
1.3.1 To describe direct blood pressure measurement
1.3.2 To illustrate indirect blood pressure measurement

**1.4.0 To Understand Blood flow meters**
1.4.1 To explain the working principle of electromagnetic blood flow meter
1.4.2 To describe the principle of operation of ultrasonic blood flow meter

## <u>MODULE II</u> ECG, EMG AND EEG MEASUREMENT

**2.1.0 To Understand the ECG measurement**
2.1.1 To explain ECG wave form.
2.1.2 To describe Einthoven's Triangle
2.1.3 To explain briefly the different electrodes used for ECG measurement.
2.1.4 To explain the unipolar lead , bipolar lead and chest lead system.
2.1.5 To describe the block diagram of an ECG machine.

**2.2.0 To Understand the electrical activity of muscles**
2.2.1 Draw the typical EMG waveform
2.2.2 To explain briefly the different electrodes used for EMG measurement
2.2.3 To describe the block diagram of an EMG machine

**2.3.0 To Understand the electrical activity of Brain.**
2.3.1 To explain the different frequency regions of EEG waveform
2.3.2 To describe briefly the different electrodes used for EEG measurement
2.3.3 To explain the 10 – 20 electrode Lead system.
2.3.4 To describe the block diagram of an EEG machine.

## <u>MODULE III</u> THERAPEUTIC AND ANALYTIC INSTRUMENTS

**3.1.0 To Understand the different types of therapeutic instruments**
3.1.1 To State the need of pacemakers
3.1.2 To Classify different types of pacemakers
3.1.3 To Compare the implantable pacemakers and external pacemakers
3.1.4 To Draw the block diagram of a ventricular synchronous demand pacemaker and to explain its operation.
3.1.5 To State the need of defibrillators.
3.1.6 To describe ac defibrillators and dc defibrillators.
3.1.7 To explain the functions of haemodialysis machine.
3.1.8 To State the use of respirators.

3.1.9 To State the need of ventilator.
3.1.10 To List the different types of diathermy equipments.
3.1.11 To explain the working of short wave diathermy unit
3.1.12 To List the advantages and disadvantages of short wave diathermy treatment.
3.1.13 To explain the working of Ultrasonic diathermy unit .

**3.2.0 To Understand the necessity of blood cell counting**
3.2.1 To explain the working of electrical conductivity blood cell counter

**MODULE IV IMAGING SYSTEMS, BIO TELEMETRY & SAFETY PRECAUTIONS**

**4.1.0 To understand the methods of imaging systems**
4.1.1 To explain the construction and operation of an X-ray machine with block diagram
4.1.2 To describe the working principle of CAT scanner
4.1.3 To explain the working principle of an ultrasonic imaging system (A Scan, B Scan and M Scan)
4.1.4 To describe the working principle of nuclear magnetic resonance imaging system (Basic NMR Components)

**4.2.0 To Understand Bio telemetry**
4.2.1 To state the need of Bio telemetry
4.2.2 To explain the block diagram of biotelemetry system
4.2.3 To describe the applications of biotelemetry – telemedicine, and video conferencing

**4.3.0 Understand Safety precautions in biomedical field**
4.3.1 To list the effect of electricity, electromagnetic radiation & magnetism in the Human body
4.3.2 To state the precautions to be taken while handling biomedical instruments
4.3.3 To list the electrical safety considerations with respect to machine operators and patients
4.3.4 To define Macro shock – Micro shock

## CONTENT DETAILS

### MODULE I

Bio-electricity of a cell- Resting and action potentials of a cell- Sodium pump and transmission of impulses.
Criteria for selecting biomedical transducers- Working principle of optical fibre temperature sensors- photo electric pulse transducers- piezo-electric arterial pulse receptor- strain gauge type respiration sensor-Direct blood pressure measurement-Indirect blood pressure measurement- electromagnetic blood flow meter- ultrasonic blood flow meter .

### MODULE II

ECG wave form- Einthovens Triangle- Different electrodes used for ECG measurement- Unipolar lead - Bipolar lead and Chest lead system- Block diagram of an ECG machine- EMG waveform-Different electrodes used for EMG measurement- Block diagram of an EMG machine- Different frequency regions of EEG waveform- Different electrodes used for EEG measurement- The 10 – 20 electrode Lead system- Block diagram of an EEG machine.

**MODULE III**

pacemakers - types of pacemakers- Implantable pacemakers and external pacemakers- ventricular synchronous demand pacemaker - defibrillators- ac defibrillators and dc defibrillators- haemodialysis machine -Respirators- ventilator- diathermy equipments - short wave diathermy unit- Advantages and disadvantages of short wave diathermy treatment- Ultrasonic diathermy unit - electrical conductivity blood cell counter.

**MODULE IV**

X-ray machine block diagram - CAT Scanner- ultrasonic imaging system (A Scan, B Scan and M Scan)- nuclear magnetic resonance imaging system (Basic NMR Components)- Bio telemetry- Block diagram of biotelemetry system - Applications of biotelemetry – Telemedicine, and video conferencing - Effect of electricity , electromagnetic radiation and magnetism in the Human body-Precautions to be taken while handling biomedical instruments- Electrical safety considerations with respect to machine operators and patients- Macro shock – Micro shock

**REFERENCES**

1. R. S. Kandpur - Hand Book of Biomedical Instrumentation – Tata MacGraw-Hill
2. Leslie Cromwell & Free. J.Weibell - Biomedical Instrumentation and Measurements _ Prentice-Hall
3. John. G. Webster - Medical Instrumentation - Wiley India
4. Dr. Arumugham - Biomedical Instruments –
5. Joseph Carr & Joseph Brown - Introduction to Biomedical equipment Technology – Pearson